Audit of the SV efficiency proof, round 4: after the R1–R2 corrections

Source audited: `sv_efficiency_proof.md` (the source of `sv_efficiency_proof.tex`), rendered as `sv_efficiency_proof_md.pdf`, at the state of commit "Fix audit round-3 issues R1 and R2". Earlier rounds: 1, 2, 3. Not refereed.

Part A. Proof-check

A1. Scope

This round re-checks the passages changed in round 3: the §1 quadrature-defect block, Lemma H(iii), §9, Theorem 4.3, the §10 split, Proposition 5.2, Lemma V and §8. It also does a third full pass focused on the interplay of the test spaces (A2 against the §1 local dual norms) and on boundary patches.

A2. Step table

tep	tool	hypothesis	constant	status	
1, $\delta^{\rm quad}$ efinition	two quadrature rules on $ \mathrm{Res}_A(\psi_a)$; the stabilisation is exact (edge form $=(A\nabla u_h,\nabla v)$, H6)	compatible patches only ($\delta_a=0$ elsewhere)	—	OK. Exactness conditions re-checked against the integrand degrees with P1 tests: $\varepsilon\le4$, $f,b\le3$, $c\le2$ for the degree-4 assembly rule; the degree-6 estimator rule is exact on the same data	
1, sign of $\delta^{\rm lg}$	$\delta^{\rm alg}_a=-r_a$	—	—	OK (matches `sv_algebraic`)	
emma H(iii)	$\ell(\psi_a)=\delta^{\rm alg}_a+\delta^{\rm quad}_a$; H(i), H(ii)	A5 for the global part	C_Π	OK	
9 nodal moments	same	—	—	OK	
hm 4.3, parenthetical	"$\eta_{\rm def,A}\le C_\Pi\,\mathrm{osc}_q$ at the discrete solution"	**needs A5**, but the same paragraph says "this direction needs no A5"	C_Π	**S1** (wording)	
10 split	$\mathrm{Res}_A(\psi_a) =-r_a+\delta^{\rm quad}_a$; $\eta_{\rm disc }:=\kappa\eta_A+\eta_S+ \mathrm{bd}-\eta_{\rm alg}$	—	exact	OK. $\eta_{\rm disc}\ge0$ is not guaranteed in principle (it is if $\eta_{\rm def}(\psi^{\rm quad})$ is below the patch part of η_A), but Proposition 5 never needs it: part 3 presupposes $\eta_{\rm alg}\le\gamma\eta_{\rm disc}$	
rop. 5.2	$\|\ell^{\rm alg}_{\rm def}\|_*\le\|\ell_{\rm def}\|_*+\|\psi^{\rm quad}\|_*\le C_I\|\mathrm{ Res}_A\|_*+\mathrm{osc} _q$; A5	A1–A6	κC_Π	OK	
emma V, diffusion part	$(Q_K^{\rm asm}-Q_K^{\rm est})(\nabla u_h\cdot\nabla\psi_a)	_K$	—	—	OK (sign checked: δ contains $-\int D\nabla u_h\cdot\nabla\psi_a$)
2 vs the §1 local norms	A2's test space on free patches is $\{w=0\text{ on }\partial\omega_a\cap\ Gamma_D\}$ (dual to a free normal flux on Γ_D); the local dual norms use H^1_*	—	—	OK. A2's space enters only through Poincaré/Friedrichs in Lemma G; the efficiency direction uses bubbles, which lie in both spaces	
emma G case 2, free atches	Friedrichs on ω_a with $w=0$ on a Dirichlet face	the free patch has a Dirichlet face of positive measure (by definition)	$C(d,\vartheta)$	OK	

tep	tool	hypothesis	constant	status	
emma A vs SV20's ecessity statement	the cut-off min	—	—	**S2**: see below	
3 [AV11]	—	—	—	**S3**: the venue is now verifiable	

A3. Verdict

- **All results are correct**, given A2 (Neumann version, [EV15]) and A5 (Licht 2019, abstract-level). R1 and R2 are closed.
- Three minor points remain, S1–S3. None affects a statement's truth or a computed bound.

A4. Remaining points

S1 (Theorem 4.3, wording). The parenthetical "≤ C_Π osc_q" uses A5, while the proof says "this direction needs no A5". **Fix:** add "(with A5)" to the parenthetical. The A5-free statement is $|\ell_{\rm def}|_*=\mathrm{osc}_q$ at the discrete solution, and $\eta_{\rm def,A}$ is only an upper bound of that.

S2 (reconciling the weight-free construction with SV20).

- **The apparent conflict.** Smears–Vohralík state that the weights in the flux construction are necessary for robustness of any flux-equilibration estimate "that does not employ submeshes or estimators combination". The galerkit patch problems carry no weights (Lemma G, ◆), so a referee will ask.
- **Why there is none.** The answer is Lemma A: the evaluation $\min\{\varepsilon^{-1/2}|\tau|_K,\ \sum_Fc_F|\tau\cdot n|_F+m_K|\mathrm{div}\,\tau|_K\}$ is precisely an estimator combination. Its second entry carries the weight $\theta_K^{1/2}$, so the result falls under SV20's exception and does not contradict it.
- **Fix.** Add a remark after Lemma A, or in §0 item 1, saying this. The proof itself is unchanged. The best-approximation trick (Lemma G case 1) is what makes the unweighted minimiser good enough for the combined evaluation.

S3 ([AV11] venue). It is now verified (Strathclyde and ASEP records): M. Ainsworth, T. Vejchodský, Fully computable robust a posteriori error bounds for singularly perturbed reaction–diffusion problems, Numer. Math. 119(2) (2011) 219–243, doi 10.1007/s00211-011-0384-1. **Fix:** replace "venue from memory".

Status of the fixes (18 Oct 2026)

- **S1:** fixed. Theorem 4.3 now separates the A5-free identity ‖ℓ_def‖_* = osc_q from the bound η_def,A ≤ C_Π osc_q, which needs A5.
- **S2:** fixed. New Remark A.1 after Lemma A: the cut-off evaluation is an estimator combination, so SV20's necessity of weights does not apply. The best-approximation trick explains why the weight-free minimiser suffices, and removing the min would lose robustness. §0 item 1 points to it.
- **S3:** fixed. [AV11] is given with its verified venue: Numer. Math. 119(2) (2011) 219–243, doi 10.1007/s00211-011-0384-1.

A5. Optimality (changes since round 3)

- **No change** to the hypotheses-used list or to the sharpness assessment.
- **The c_b-free track** (cellwise σ, adapting [AV14]) remains the most useful extension.
- **Strongest conclusion.** The weight-free construction is the stronger form: the same robustness as SV20 with simpler patch problems. With the S2 remark it can be presented as such.

Part B. Literature scan (round 4)

B1. Answer

There is no change in novelty or track. One new preprint needs a careful read because it bears on Theorem 6.

B2. New items

result	scheme	estimate	hypotheses	mechanism	relevance
He, Equilibrated averaging residual method (EARM / ON-EARM), arXiv 2601.01585 (Jan 2026)	conforming (ON-EARM), elliptic interface problems, 2D	robust a posteriori estimator **for the recovered flux**, robust in coefficient jumps	2D; per the abstract	equilibrated averaging + null-space elimination	**[verify]** Its claimed jump-robustness sits uneasily with Li 2026 (no local estimator is robust in the contrast for conforming approximations). The likely reconciliation: robustness of the recovered-flux error rather than the energy error, or a non-local step. Read before citing either in §11
Chaumont-Frelet, Dong, Gantner, Vohralík, arXiv 2603.08887 (2026)	h-adaptive FEM, Poisson	p-robust error contraction under equilibrated-flux marking	verifiable criterion	equilibrated fluxes	context for the adaptive study; Poisson only
Ainsworth–Vejchodský, Numer. Math. 119 (2011) 219–243	conforming P1, reaction–diffusion	first fully computable robust bounds	constant reaction, 2D	Ainsworth–Oden fluxes, local problems not solved exactly	verified (S3)
Smears–Vohralík (SV20), necessity of weights	equilibration, reaction–diffusion	weights necessary unless submeshes or estimator combination	—	—	**S2**

Unchanged: Li 2026, Licht 2019 (abstract only), AV14, Ahmed–Jha 2026, Jha 2021/2024, Verfürth 2005, Cai & Cai 2025, Eigel–Merdon 2016, ESV10, ABR17, Ern–Vohralík 2013, HPSV21. No 2026 work on equilibrated estimators for algebraic flux correction was found.

B3. Open items, ranked by novelty-to-effort

1. **S1–S3:** text only.
2. **Read EARM (arXiv 2601.01585) against Li (arXiv 2609.27817)** before citing either in §11. Low effort.
3. **The c_b-free track with a cellwise σ** (adapting AV14). Medium effort.
4. **A5:** read Licht 2019, Theorem 1. Low effort; needs network access.
5. **The stopped iterate against the discrete solution.** Open, high risk.

B4. References (new in round 4)

- M. Ainsworth, T. Vejchodský, Numer. Math. 119 (2011) 219–243: Strathprints, ASEP, NCMM, preprint. [abstract]
- EARM / ON-EARM, arXiv 2601.01585. [abstract]
- T. Chaumont-Frelet, Z. Dong, G. Gantner, M. Vohralík, arXiv 2603.08887. [abstract]
- I. Smears, M. Vohralík, arXiv 1812.06678 (the necessity of weights). [read]
- Ainsworth–Vejchodský 2014, export.arxiv.org. [abstract]